

LitSense Weekly Progress Report

November 24, 2025

This Week's Work: Built 7 AI agents that automatically detect when researchers struggle with papers and connect them with help from AI or other researchers

What Works Now: System monitors reading behavior, detects confusion after 3 highlights, matches struggling researchers with helpers who understand the section, and opens a help panel with AI chat and peer connections

Module Progress Status

#	Module Name	Status	%
2	Interaction Analysis	Complete	100%
3	Discussion Analysis	In Progress	90%
4	Role Assignment	Complete	100%
5	Matching System	Complete	100%
1,6-10	Remaining Modules	Pending	0%

Multi-Agent System: What Each Agent Does

Monitoring & Detection

- Agent 1:** Watches researchers read - checks every 5 seconds if they're struggling by tracking confusion highlights, time spent, and eye movements
- Agent 4:** Analyzes highlights - spots patterns like "3 people highlighted the same paragraph" and suggests they should talk
- Agent 5:** Records everything - keeps a log of struggles, breakthroughs, and who helped who for later analysis

Help & Communication

- Agent 2:** Finds the best person to help - looks at who understands the section well and is available, gives each match a score out of 100
- Agent 3:** Sets up conversations - opens chat windows between researchers and suggests what to talk about based on the confusion
- Agent 6:** Explains concepts - uses GPT-4 to give simple explanations when researchers ask questions
- Agent 7:** Sends smart alerts - shows notifications like "This section is tricky, want AI help?" or "Demo User can help you with this"

AI Coordination Core (The "Brain")

- Acts as the central controller - when something happens (like a researcher struggles), it decides which agents should respond
- Prevents chaos - makes sure agents don't send conflicting messages or duplicate notifications
- Keeps everyone in sync - maintains a central record of who's doing what and ensures agents work together smoothly

Real-Time Collaboration: What Works Now

Technology

- Uses Socket.io (like WhatsApp - instant messaging between browsers)
- Each paper gets its own "room" where collaborators join
- Messages delivered in under 100 milliseconds
- Auto-reconnects if internet drops

Live Features

- Person A highlights text → Person B sees it instantly
- Assign sections → everyone sees who's responsible

- Green/gray dots show who's online
- Notifications sent to specific people (not spam)
- Chat messages deliver instantly

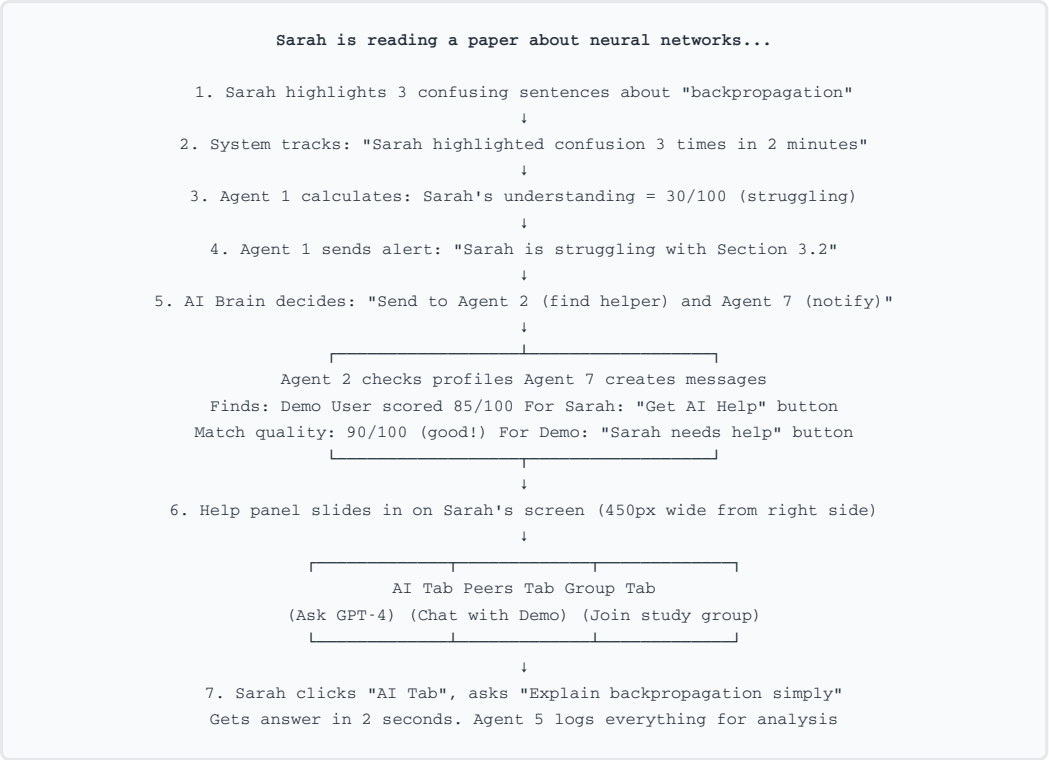
Problems Solved This Week

Challenge	Solution Implemented	Status
Notifications going to all users instead of targeted ones	Added strict filtering - each notification tagged with specific user ID	90% Fixed
User lists showing different people on different browsers	Unified to single Socket.io server, eliminated duplicate users	Resolved
Agents triggering conflicting actions	Built AI Coordination Core to route events and prevent conflicts	Resolved
Agent 2 couldn't find any peer matches (showing 0 peers)	Added auto-registration when users join + section tracking	Resolved

How The System Works: Real Example

November 24, 2025

Complete Workflow (Actual Scenario from Testing)



Data Collection: What We Track and Why

Data Type	How We Collect It	Why It Matters
Eye Movements	WebGazer.js tracks gaze position every 100ms using webcam	Shows where researchers look - erratic patterns indicate confusion
Highlights	Researchers mark text as "confused" or "understood"	Direct signal of comprehension - most reliable indicator
Time Spent	Timestamps when entering/exiting each section	Spending 2x normal time suggests difficulty
User Actions	Clicks, scrolling, section revisits	Going back to same section means they didn't understand first time
Collaboration	Chat messages, who helped who, help requests	Track effectiveness - did help actually work?

AI Integration: How GPT-4 Helps

Component	How It Works	What It Does
AI Tutor Endpoint	Dedicated API route using OpenAI GPT-4	Gives quick, simple explanations for confused text
Context Injection	Automatically includes confused highlights in the question	AI knows exactly what researcher is confused about
Conversation Memory	Keeps track of previous questions and answers	Can have back-and-forth conversation, not just one-shot Q&A
Tutor Personality	Custom prompts that make AI act like a patient teacher	Explanations are encouraging, simple, not intimidating

Next Two Weeks: Planned Work

Period	What I'll Build
Week 1	Module 1 (Reflection Analysis): Let researchers highlight whole paragraphs and track quality of their notes

Module 6 (Fact-checking): Connect to APIs that verify claims in papers (check if statistics are correct)

Module 3 (Finish): Complete the peer-to-peer chat with full Socket.io integration

Week 2

Module 7 (Related Work): Automatically find similar papers researchers should read

Module 8 (Summary Builder): Generate automatic summaries of each section